

D700

Early Mathematics Methods and Interventions

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A four-year-old in a play-based preschool with a large block corner touches each of six toy bears while saying “1, 2, 3, 4, 5, 6.” The bears are bright yellow. When the teacher asks, “How many bears?” the child starts counting again from 1. Which principle does the child not yet show?

- A. Stable order
- B. Conservation of number
- C. One-to-one correspondence
- D. Cardinality

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which behavior shows that a child understands cardinality?

- A. The child counts the cars again from the other end to check.
- B. The child recites 1 to 20 in order.
- C. The child touches each car exactly once.
- D. After counting seven cars, the child says “seven” when asked how many, without recounting.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

During a quick-image game in a kindergarten classroom decorated for fall, a teacher flashes a card with two rows of three dots for two seconds. A child who recently learned to tie her shoes says, “Six, because three and three.” What is the child showing?

- A. Conceptual subitizing
- B. Rote counting
- C. Perceptual subitizing
- D. Counting all by ones

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A first grader who loves dinosaurs and is always first in line can write $7 - 3 = 4$ but cannot explain or show what the numbers mean. The class is using a new textbook series. Using the CRA sequence, what should the teacher do?

- A. Have the child act out the problem with counters, then draw it, then connect it to the equation.
- B. Use dinosaur stickers as rewards for correct answers.
- C. Start with drawings and never use counters, since counters are for younger children.
- D. Give more practice with written equations.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A teacher has students solve problems with base-ten blocks and then sketch the blocks. What is the next step in the CRA sequence?

- A. Skip to timed tests.
- B. Connect the sketches to written numerals and equations.
- C. Return to the blocks.
- D. Have students act out the problem with their bodies.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A second grader, who takes the bus home and has a younger brother, solves $7 + 8$ by saying, “ $7 + 7$ is 14, and one more is 15.” On last week’s timed quiz, she finished only half the problems. How should the teacher view her fluency?

- A. She is not fluent because she did not finish the timed quiz.
- B. She should stop using strategies and memorize facts.
- C. She is counting on by ones.
- D. She is using an efficient derived-fact strategy that builds real fluency.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A teacher wants students to use the make-ten strategy for $9 + 6$. Which student reasoning shows it?

- A.** “9, 10, 11, 12, 13, 14, 15.”
- B.** “ $6 + 6$ is 12, plus 3 more is 15.”
- C.** “Take 1 from the 6 to make 10, then $10 + 5 = 15$.”
- D.** “ $9 + 6$ is 16, because $10 + 6$ is 16.”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A word problem from a colorful workbook reads: “Some birds were on a fence. 3 more birds came. Now there are 8 birds.” The workbook costs twelve dollars. How many birds were on the fence at first?

- A.** 12
- B.** 11
- C.** 5
- D.** 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

“Maya had 9 stickers. She gave some to her friend. Now she has 4.” What type of problem is this?

- A.** Compare, difference unknown
- B.** Separate, change unknown
- C.** Join, start unknown
- D.** Separate, result unknown

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A teacher makes two rows of seven buttons, then spreads one row out. A four-year-old who is new to the class and speaks two languages says the longer row has more. What does this show?

- A.** The child has not yet developed conservation of number, which is typical at this age.
- B.** The child has a learning disability.
- C.** The child misunderstood because of language.
- D.** The child cannot count to seven.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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11.

A kindergartner scores well below benchmark on a fall number-sense screener that checks number identification and which of two numbers is bigger. The child’s family recently moved from another state, and the child is cheerful in class. What is the best next step?

- A.** Wait until spring to see if the child catches up.
- B.** Send home extra worksheets.
- C.** Give explicit, systematic small-group instruction with visual models and monitor progress often.
- D.** Assume the low score is due to the move and ignore it.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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12.

A first grader, using a set of ten frames the school bought last year, shows 34 with three full ten frames and four more counters. Asked how many, he counts every counter one at a time. His handwriting is very neat. What does this suggest?

- A.** He lacks one-to-one correspondence.
- B.** He has not yet unitized tens; he needs practice seeing a full frame as one ten.
- C.** His neat handwriting shows he is ready for abstract place value.
- D.** He understands place value because he built 34 correctly.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A four-year-old in a play-based preschool with a large block corner touches each of six toy bears while saying “1, 2, 3, 4, 5, 6.” ~~The bears are bright yellow.~~ When the teacher asks, “How many bears?” the child starts counting again from 1. Which principle does the child not yet show?

Target word: missing principle **Decides:** Recounting instead of answering “6” shows no cardinality.

A	Dropped element	The child said the numbers in the right order.
B	Adjacent category	Conservation involves rearranged objects, which did not happen.
C	Confusion set	The child touched each bear once, showing one-to-one.
D	Correct	The child does not yet know the last number said names the amount.

2. Answer D Concept 1 • transfer

Which behavior shows that a child understands cardinality?

Target word: pick evidence of the principle **Decides:** Answering “how many” with the last number counted.

A	Inverted operation	Recounting is what a child without cardinality does.
B	Adjacent category	Shows stable order, not cardinality.
C	Confusion set	Shows one-to-one correspondence.
D	Correct	Using the last count word as the amount is cardinality.

3. Answer A Concept 2

During a quick-image game in a kindergarten classroom ~~decorated for fall~~, a teacher flashes a card with two rows of three dots for two seconds. A child ~~who recently learned to tie her shoes~~ says, “Six, because three and three.” What is the child showing?

Target word: name the skill **Decides:** Seeing two parts and combining them.

A	Correct	She saw parts (3 and 3) and combined them without counting by ones.
B	Adjacent category	Rote counting is reciting number words.
C	Confusion set	Perceptual is seeing a small amount at once without combining parts.
D	Inverted operation	Two seconds is too short to count, and she named parts.

4. Answer A Concept 3

A first grader ~~who loves dinosaurs and is always first in line~~ can write $7 - 3 = 4$ but cannot explain or show what the numbers mean. ~~The class is using a new textbook series.~~ Using the CRA sequence, what should the teacher do?

Target word: next step **Decides:** Abstract symbols without understanding means go back to concrete.

A	Correct	Return to concrete and build up to abstract.
B	Camouflage	Rewards do not build meaning.
C	Absolute language	Concrete materials are fine at any age when understanding is missing.
D	Dropped element	Stays at the abstract stage and skips the missing understanding.

5. Answer B Concept 3 • transfer

A teacher has students solve problems with base-ten blocks and then sketch the blocks. What is the next step in the CRA sequence?

Target word: next stage **Decides:** Concrete, then representational, then abstract.

A	Confusion set	Timed tests are not a stage in CRA.
B	Correct	After representational comes abstract symbols.
C	Inverted operation	Moves backward when students are progressing.
D	Adjacent category	Another concrete step, not the next stage.

6. Answer D Concept 4

A second grader, ~~who takes the bus home and has a younger brother~~, solves $7 + 8$ by saying, “ $7 + 7$ is 14, and one more is 15.” On last week’s timed quiz, she finished only half the problems. How should the teacher view her fluency?

Target word: judge fluency **Decides:** An accurate, efficient reasoning strategy (doubles plus one).

A	Adjacent category	Speed is one part; timed results are not the definition of fluency.
B	Inverted operation	Strategies lead to recall; discarding them reverses the path.
C	Confusion set	She used a double plus one, not counting on.
D	Correct	Fluency includes accuracy, efficiency, and flexibility.

7. Answer C Concept 4 • transfer

A teacher wants students to use the make-ten strategy for $9 + 6$. Which student reasoning shows it?

Target word: identify the strategy **Decides:** Move 1 from 6 to make 10.

A	Adjacent category	Counting on by ones.
B	Confusion set	That is a doubles strategy.
C	Correct	Decomposes to make a ten.
D	Dropped element	Rounds 9 to 10 but forgets to take the 1 back.

8. Answer C Concept 5

A word problem ~~from a colorful workbook~~ reads: “Some birds were on a fence. 3 more birds came. Now there are 8 birds.” ~~The workbook costs twelve dollars.~~ How many birds were on the fence at first?

Target word: solve a start-unknown problem **Decides:** $? + 3 = 8$.

A	Camouflage	The workbook’s price.
B	Inverted operation	Adds because of the key word “more.”
C	Correct	$8 - 3 = 5$.
D	Wrong stem number	The number that joined, not the start.

9. Answer B Concept 5 · transfer

"Maya had 9 stickers. She gave some to her friend. Now she has 4." What type of problem is this?

Target word: classify the structure **Decides:** Taking away with the amount taken missing.

A	Adjacent category	No two sets are compared; one set changes.
B	Correct	$9 - ? = 4$: the change is missing.
C	Inverted operation	Giving away is separating, and the start (9) is known.
D	Dropped element	The result (4) is given.

10. Answer A Concept 6

A teacher makes two rows of seven buttons, then spreads one row out. A four-year-old ~~who is new to the class and speaks two languages~~ says the longer row has more. What does this show?

Target word: interpret the response **Decides:** Judging amount by length: not yet conserving.

A	Correct	Preoperational children often judge by length.
B	Absolute language	This response is developmentally typical.
C	Camouflage	Bilingualism is not what the response shows.
D	Adjacent category	Counting was not tested; conservation was.

11. Answer C Concept 7

A kindergartner scores well below benchmark on a fall number-sense screener that checks number identification and which of two numbers is bigger. ~~The child's family recently moved from another state, and the child is cheerful in class.~~ What is the best next step?

Target word: response to screening **Decides:** Low screener score calls for targeted, explicit intervention with monitoring.

A	Dropped element	Drops early intervention.
B	Adjacent category	Practice without explicit teaching does not close the gap.
C	Correct	That is the evidence-based response to early risk.
D	Camouflage	The move does not cancel the screening result.

12. Answer B Concept 8

A first grader, using a set of ten frames ~~the school bought last year~~, shows 34 with three full ten frames and four more counters. Asked how many, he counts every counter one at a time. ~~His handwriting is very neat.~~ What does this suggest?

Target word: interpret counting by ones **Decides:** He does not yet treat a full frame as one ten.

A	Confusion set	He counted each counter once, which shows one-to-one.
B	Correct	Counting by ones when groups exist shows tens are not yet units.
C	Camouflage	Handwriting does not show number understanding.
D	Inverted operation	Building the model does not mean he sees tens as units.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Counting principles and cardinality	Counting needs stable order, one-to-one correspondence, and cardinality (the last number said tells how many).	Assuming a child who recites numbers in order and touches each object understands how many.	A child recounts when asked “How many?”
2	Perceptual vs. conceptual subitizing	Perceptual subitizing is instantly seeing a small amount (up to about 4-5) without counting; conceptual subitizing sees parts and combines them (3 and 3 make 6).	Calling any quick answer counting, or calling part-whole recognition perceptual.	Quick-flash dot cards or dice.
3	Concrete-Representational-Abstract	Build understanding with objects first, then drawings or models, then symbols; move back a stage when a student is stuck.	Starting with symbols or treating manipulatives as only for struggling students.	Blocks, then pictures, then numerals.
4	Fact fluency through strategies	Fluency means accurate, efficient, and flexible; it grows from reasoning strategies (make ten, doubles, doubles plus one), not timed drills alone.	Equating fluency with speed on timed tests.	A student’s way of solving a basic fact.
5	Addition and subtraction problem types	Join, separate, part-part-whole, and compare problems can have the result, change, or start unknown; start-unknown problems are the hardest for young children.	Teaching key words (“more means add”) instead of the problem’s structure.	Which quantity is missing in the story.
6	Conservation of number	Children who conserve know a quantity stays the same when objects are spread out or pushed together; preoperational children often think the longer row has more.	Treating the non-conserving answer as carelessness.	Two equal rows, one spread out.
7	Screening and explicit intervention	Early numeracy screeners (number identification, quantity discrimination, missing number) flag risk; effective intervention is explicit and systematic, with modeling, guided practice, visual models, and frequent checks.	Waiting to see if the child catches up, or only giving more worksheets.	A young child scoring low on a number-sense screener.
8	Unitizing with ten frames and base-ten models	Understanding place value means seeing ten ones as one ten (unitizing); ten frames and bundles build this.	Thinking a child who writes 34 correctly understands it as 3 tens and 4 ones.	A child counting by ones when groups of ten are available.

Course scope (public description): Early Mathematics Methods and Interventions is a comprehensive course designed for educators focused on advancing mathematics education in the early elementary grades. This course emphasizes innovative and research-based teaching methods for developing mathematical understanding in

young children, particularly in the crucial developmental stages in PK through the early elementary grades. Participants will explore a variety of instructional strategies to support conceptual understanding and procedural fluency in areas such as number sense, basic operations, and early problem-solving skills. A significant aspect of the course involves identifying and addressing learning needs through targeted interventions, personalized instruction, and the use of manipulatives and digital tools. The curriculum also highlights the importance of creating an engaging and inclusive learning environment that encourages exploration and curiosity in mathematics. By integrating assessment techniques and thoughtful learning experiences, this course aims to provide educators with the skills to effectively nurture early mathematical abilities and lay a strong foundation for future academic success in mathematics.